

Michael Penhallegon

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Summary:

A data-focused software engineer with a focus in API development, building data pipelines, and ETL (extract, transform, and load) processes. Has a diverse set of skills from application development to data analytics. Has utilized mixtures of skill-sets to develop projects improving the data utilization and access of data to answer research and business questions. Love new challenges and investigating applications of software and data engineering in new contexts.

- 6+ years of experience with Python and advanced knowledge of R, and SQL in production development environments
- Skilled in understanding and designing ETL pipelines to wrangle and process fragmented data using a variety of technologies such as AWS Lambda, PySpark, Airflow, step functions and Athena/Presto
- Collaborative leadership connecting business stakeholders with data solutions
- DevOps skill-sets utilizing container images, S3, Apache Airflow, Postgresql, Kubernetes, and Docker

Experience:

Crown Castle
Software Engineer II

Seattle, WA
April 2023-April 2025

- Developed, troubleshooted and maintained a collection of Python based microservices (from gRPC, RESTful or general) for surfacing, analyzing and integrating data from telecom device inventory and other systems.
- Developed, troubleshooted and maintained a collection of PySpark based data pipelines.
- Lead and helped expand integration testing and unit testing into codebases.
- Integrated and used monitoring for monitoring, alerts and dashboarding
- Developed Grafana Dashboard to provide graphical user interfaces to data products and services.
- Wrote product plans and other project/product documentation to improve planning and support of projects maintained by the team.
- Used agile and scrum development practices to collaborate within the team to ensure delivery of projects

DomainTools
Software Engineer II

Seattle, WA
July 2021-August 2022

- Developed, troubleshooted and maintained multiple Python-based FastAPI RESTful microservices
- Used unit and integration testing patterns to maintain quality engineered code bases
- Integrated and configured Prometheus monitoring, alerting and Grafana dashboards
- Troubleshooted bugs and data quality issues from front-end down into back-end and data storage systems
- Wrote and developed complex PostgreSQL function and stored procedure against large datasets to transform data and maintain good data warehousing metadata
- Developed integrated data reporting from multiple microservices and databases to provide operational usage statistics and analytics to management and stakeholders across a hybrid on-prem/cloud environment using S3 and AWS Lambda
- Used agile and scrum development practices to triage and collaborate within a team and across teams

Bristol Myers Squibb

Data Engineer

Seattle, WA

August 2018-June 2021

- Collaborated and developed software and data solutions, focussing on backend data engineering
- Streamlined data ingestion and delivery, reduced time for data transfers utilizing automation, and optimized data delivery from days to hours, leading to cleaned data stored in S3 and accessed with AWS Athena
- Led and contributed to the collection of requirements and creation of project plans with multiple partners and collaborators to build data, software, and visualization tools.
- Developed visualizations, web applications, and RESTful APIs; using R/Python
- Developed and maintained multiple data pipelines utilizing Airflow and containers (with Docker or Amazon ECS) to load, transform, and ingest batches of data into OLAP databases (primarily Postgres and Redshift).
- Developed over five different internal packages and libraries for team members to utilize upstream data and RESTful APIs in consistent and re-usable ways

Fred Hutchinson Cancer Research Center

Research Intern

Seattle, WA

June 2017-June 2018

- Placed in the HIV Vaccine Trial Network Core Lab; Inclusive of two internship terms (Summer 2017 & Spring 2018) with complimentary work between terms.
- Completed thorough literature review and prepared results for publication in an academic journal.
- Contributed to a qualitative research project investigating HIV/AIDS stigma among HIV+ and HIV- men
- Developed and published an R package, to linearly interpolate Census-data to generate health metric values.

Education:

The Evergreen State College

September 2015-June 2018

Bachelor of Arts, emphasis in Computer Science and Public Health Studies

Volunteer Experience:

Sisters of Perpetual Indulgence, Abbey of St Joan, Secretary, September 2019-Present

2021 King-Snohomish-Island Counties Ryan White RFP Review Committee, Member, March 2021

2019 King-Snohomish-Island Counties Ryan White RFP Review Committee, Member, December 2018